AICTE · SAGE PLATFORM DOCUMENTATION

Windows OS Setup & Execution Guide

Complete step-by-step instructions for cloning, configuring, and executing SAGE on Microsoft Windows 10/11.

Repository: https://github.com/Shreyash414/SAGE.git

Target OS: Windows 10 / Windows 11 (64-bit)

Stack: Java 21 LTS · React 19 · PostgreSQL 18 · MinIO

Default Shell: Windows Command Prompt (CMD) or PowerShell

01 System Prerequisites on Windows

Before cloning, ensure the following core software is installed on your Windows machine:

Tool / Runtime	Required Version	Installation Source	Verification Command
Git for Windows	2.30+	git-scm.com	`git --version`
Java Development Kit (JDK)	JDK 21 LTS	Eclipse Temurin 21 or Microsoft OpenJDK	`java -version`
Node.js & npm	v20.x or v22.x LTS	nodejs.org (includes npm 10+)	`node -v && npm -v`
Docker Desktop	v4.25+ (WSL 2 backend)	docker.com	`docker compose version`

CRITICAL: Make sure **Docker Desktop** is opened and actively running in your system tray before executing container commands.

02 Clone the Repository

Open **Command Prompt (cmd)** or **PowerShell** as a regular user, navigate to your desired directory, and clone the repository:

```
:: Navigate to your working workspace
cd %USERPROFILE%\Projects

:: Clone the official GitHub repository
git clone https://github.com/Shreyash414/SAGE.git

:: Enter the project root
cd SAGE
```

03 Environment Configuration (.env)

The project provides an automated configuration template. Create your local `.env` file:

```
:: Command Prompt (CMD)
copy .env.example .env

:: OR in Windows PowerShell
Copy-Item .env.example .env
```

The default `.env` values are pre-configured to work immediately with the included Docker services:

Parameter	Default Value	Description
`SERVER_PORT`	`8080`	Backend REST API server port
`DB_PORT`	`5432`	PostgreSQL database port
`DB_NAME` / `DB_USER`	`aicte_platform_db` / `aicte_app_user`	Application database credentials
`MINIO_ENDPOINT`	`http://127.0.0.1:9000`	MinIO S3 storage API endpoint
`MINIO_CONSOLE_PORT`	`9001`	MinIO administrative web dashboard

04 Start Database & Object Storage

Run Docker Compose to launch PostgreSQL 18 and MinIO Object Storage:

```
:: Start containers in detached mode
docker compose up -d

:: Verify both containers (aicte-postgres and aicte-minio) are healthy and running
docker ps
```

> **MinIO Console:** You can access the MinIO web dashboard at `http://localhost:9001` using user: `sage_minio_admin` and password: `sage_minio_secret_2026`.

05 Start the Spring Boot Backend (Terminal 1)

Open a new Command Prompt or PowerShell in the `SAGE` directory and run the pre-configured Windows Maven wrapper (`mvnw.cmd`):

```
:: Windows Batch Wrapper automatically resolves dependencies and runs Flyway migrations
mvnw.cmd spring-boot:run
```

Upon starting, Flyway executes migrations **V1 through V6**, creating schemas, statutory document catalogues, workflow state machines, and default seed accounts. The backend is live when you see:

```
Started AictePlatformApplication in X.XXX seconds (process running for X.XXX)
```

- **REST API Root:** `http://localhost:8080`
- **Health Check Endpoint:** `http://localhost:8080/api/v1/health`
- **Swagger OpenAPI UI:** `http://localhost:8080/swagger-ui.html`

06 Start the User Portal (Terminal 2)

Open a second terminal window, navigate into the `frontend` directory, install dependencies, and launch Vite:

```
cd frontend
npm install
npm run dev
```

The User Portal is now accessible in your browser at: **`http://localhost:5173`**.

07 Start the SAGE Admin Panel (Terminal 3)

Open a third terminal window, navigate into the `admin` directory, install dependencies, and launch Vite:

```
cd admin
npm install
npm run dev
```

The Admin Panel is now accessible in your browser at: `http://localhost:5174`.

> **Note:** On Windows, always run `npm install` separately inside both `frontend` and `admin`. This installs native Windows build binaries for both frontends cleanly.

08 Seeded Demonstration Accounts

The platform comes pre-seeded with test accounts for all AICTE approval roles:

Role	Username / Email	Password	Target Portal
System Administrator	`admin@aicte.gov.in`	`Admin@123456`	Admin Panel (Port 5174)
Scrutiny Officer	`scrutiny@aicte.gov.in`	`Scrutiny@123456`	User Portal (Port 5173)
Regional Officer	`regional@aicte.gov.in`	`Regional@123456`	User Portal (Port 5173)
EVC Expert	`evc@aicte.gov.in`	`Evc@123456`	User Portal (Port 5173)
Institution Applicant	`principal@tech.edu`	`College@123456`	User Portal (Port 5173)

09 Common Windows Troubleshooting & FAQ

1. "Port 5432 is already in use" Error

- **Cause:** You already have a native PostgreSQL service installed and running on Windows.
- **Solution:** Open `services.msc` in Windows, locate the **postgresql-x64** service, right-click, and select **Stop**. Alternatively, edit `DB_PORT=5433` in your `.env` file and restart `docker compose up -d`.

2. "Cannot connect to the Docker daemon"

- **Cause:** Docker Desktop is not running or the WSL 2 integration has not initialized.
- **Solution:** Open **Docker Desktop** from the Windows Start menu. Wait until the status indicator in the bottom-left corner turns green.

3. PowerShell Script Execution Policy Error

- **Cause:** Windows PowerShell by default restricts running unsigned scripts.
- **Solution:** Run PowerShell as Administrator and execute:

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

4. 'mvnw' is not recognized as an internal or external command

- **Cause:** On Windows Command Prompt / PowerShell, Unix shell scripts without extensions do not run directly without typing the full file name.
- **Solution:** Make sure you run `mvnw.cmd spring-boot:run` with the `.cmd` extension.

5. Flyway Migration or CRLF Line Ending Issues

- **Cause:** Git for Windows may convert line endings from LF to CRLF upon checkout.
- **Solution:** If you see SQL parser errors, configure Git to preserve input line endings:

```
git config --global core.autocrlf input
```